Muhammad Owais Mushtaq

Senior Software Engineer — Tech Lead — FinTech

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Summary

Senior Software Engineer and tech lead with 4+ years building wealth-management FinTech products for US advisory firms. Hands-on across backend, frontend, data, cloud and delivery: I modernize legacy systems without big-bang risk, build new products from scratch, and keep improving what is already running. I lead a five-person team, own code reviews, merges and releases, and run an AI-first workflow that keeps the pace high and the quality bar higher. MS in Software Project Management and published researcher in AI-assisted software engineering.

Professional Experience

Matrix Systems (Pvt.) Ltd **Karachi, Pakistan**

Engineering team behind SoftPak's wealth-management products (uRebal, uRebal Hub, MARS)

Senior Software Development Engineer / Tech Lead *Jan 2026 – Present*

- Lead the modernization of **uRebal**: rebuilt its legacy Jersey backend as enterprise Spring Boot APIs that serve both the standalone product and the **uRebal Hub** SaaS platform; primary system designer and author of the core API contracts.
- Replaced hand-managed threads with **RabbitMQ** for scalable parallel processing, and moved native C++ library integration from JNI to Java's **Foreign Function & Memory (FFM) API** for safer error handling.
- Moved PostgreSQL manual scripts and triggers to **Liquibase**, shifted MySQL trigger logic into the application layer, and made the application **multi-tenant** with audit trails, Keycloak security, Swagger documentation, structured logging and environment-specific configuration.
- Introduced a **backend-for-frontend (BFF)** layer so the existing UI runs end-to-end on the new APIs and QA can sign off before direct frontend integration; wrote release notes for the uRebal Hub team covering requests, responses, database changes and API behaviour.
- Run an **AI-first workflow**: evaluated Junie, Codex, Claude, GitHub Copilot and Cursor across the team before standardizing on Claude; wrote repository guidelines for coding, code review, unit testing and AI agents, so developers build features from a Jira ticket with minimal rework.
- Own the quality gate: every pull request passes SonarCloud and Copilot review, followed by my final review of business logic, performance and consistency before I merge and release.
- Built end-to-end CI/CD for each platform (Jenkins for SVN, GitHub Actions, Bitbucket Pipelines): build, Docker image, deployment to dev, QA, UAT and production on AWS and Azure, SonarCloud and JaCoCo quality gates, and release notifications.
- Defined Technical Solution (TS) and Product Integration (PI) workflows for the company's **CMMI V3.0 Level 3** appraisal; estimate and break down work with product managers.

Software Development Engineer II *Jan 2024 – Dec 2025*

- Co-architected **Advisor Sketchbook**, a multi-tenant FinTech SaaS for financial advisors, from requirements to MVP, working directly with US clients and product managers in Scrum.
- Built complete Angular applications from Figma designs, scaling them with micro-frontends (Module Federation) and data-heavy components such as multi-level grids and dynamic charts.
- Integrated **Keycloak** for enterprise SSO and MFA using OpenID Connect (OIDC).
- Designed Bitbucket CI/CD pipelines for containerized builds and multi-environment deployments to Azure Kubernetes Service (AKS) via ACR.
- Delivered financial reporting with JasperReports; reviewed code and mentored junior developers on design and maintainability.

Software Development Engineer *Oct 2022 – Dec 2023*

- Optimized mission-critical SQL for **MARS**, cutting a rebalancing job from **10 minutes to under 1 second** across 2,000+ proposals.
- Fixed C++ calculation engines and extended a Microsoft Foundation Classes (MFC) utility with **BCP** bulk loading, moving client Excel and flat files from Azure Blob Storage into cloud SQL databases.
- Delivered Risk Management and Portfolio Accounting features end to end in Kanban; upgraded Angular 12 to 15 and introduced a shared component library (Bootstrap, jqWidgets).
- Automated deployments with GitHub Actions, improved legacy Jenkins pipelines, and redesigned Jasper-Reports for faster, cleaner financial reports.

Associate Software Development Engineer *Jul 2022 – Sep 2022*

- Developed Spring Boot REST APIs with OpenAPI documentation; maintained MS SQL integrity through incremental, rollback and stored-procedure scripts.

FAST School of Management, FAST-NUCES **Karachi, Pakistan**
Research Assistant (funded project, Dr. Nazia Nazeer) *Jan 2025 – Jan 2026*

- Planned a funded field study on rural working women in Pakistan's date palm sector and the SDGs: literature review, methodology, institutional MoUs, and a three-person field team; manuscript under review.

Folio3 Software **Karachi, Pakistan**
Software Engineering Intern *Jun 2022 – Aug 2022*

- Built a full-stack web application in a hybrid Agile cohort under senior mentorship.

FAST-NUCES **Karachi, Pakistan**
Teaching Assistant *Sep 2021 – Jun 2022*

- Supported vivas, quizzes and projects for Parallel and Distributed Computing and Digital Logic Design.

Education

National University of Computer and Emerging Sciences (FAST-NUCES), Karachi
Master of Science in Software Project Management – CGPA 4.0/4.0 *Jun 2026*

- **Thesis:** AI-Augmented Agile MVP Development for Lean Tech Startups.

Bachelor of Science in Computer Science – CGPA 3.02/4.0 *Jul 2022*

- **Final Year Project:** Vision-Based Human Activity Recognition, comparing CNN+LSTM, Inception-V3 and MobileNet on UCF-101 (A+ in FYP I and II).
- Dean's List of Honor (Spring 2022); competed in speed programming (Coder's Cup, DevDay).

Research & Publications

- **M. O. Mushtaq** (2026). AI-Augmented Agile MVP Development for Lean Tech Startups. *Research Square* preprint, doi:10.21203/rs.3.rs-10346557/v1. Under review at *Innovations in Systems and Software Engineering* (Springer).
- N. Nazeer, R. Rasiah, **M. O. Mushtaq**, T. Shahid, A. A. Sheikh (2026). Beyond aggregate scorecards: An intersectional and capability analysis of rural working women's lived experiences of sustainable development in Pakistan's date palm industry. Under review.

Selected Projects

- **Brain Tumor Detection:** custom CNN vs VGG-16 transfer learning on brain MRI scans; VGG-16 reached 97.95% test accuracy vs 89.16%.
- **ATM Cash Predictor:** next-day ATM cash demand with linear regression and XGBoost; XGBoost raised accuracy from about 50% to 85%.
- **Parallel Cryptography:** C++ and OpenMP; text split across four ciphers and encrypted in parallel.

Technical Skills

Backend & Core	Java (Spring Boot, Jersey, FFM API), Hibernate/JPA, Spring Cloud, RabbitMQ, TypeScript (Node.js, Nest.js), C++ (MFC), .NET (6, 8), Python, SQL
Frontend & UI	Angular (12–15), Module Federation, Material UI, jqWidgets, Bootstrap, Figma-to-code, responsive UI
Cloud & DevOps	AWS (EC2, ECS, ECR, EFS, S3), Azure (AKS, SQL, DevOps, File Share, Blob, Data Lake), Docker, Kubernetes, GitHub Actions, Bitbucket Pipelines, Jenkins, Liquibase
Data	MS SQL Server (query optimization, BCP), PostgreSQL, MySQL, MongoDB, Redis, schema design, database versioning, multi-tenancy
Security & Quality	Keycloak (SSO, OIDC, MFA), OpenAPI/Swagger, audit logging, SonarCloud, JaCoCo, JUnit, Mockito, JasperReports
AI & Research	Claude, Codex, GitHub Copilot, Cursor, Junie; LLMs, LangChain, prompt engineering; TensorFlow, deep learning; Design Science Research
Methods	Agile (Scrum, Kanban), CMMI V3.0 Level 3 (TS, PI), estimation and work breakdown, technical solution design
Tools	Linux, Git, SVN, Jira, Confluence, Postman, VS Code, PyCharm, PuTTY, Hangfire

Leadership & Professional Development

- **Aspire Leadership Program (2024):** global leadership training with Harvard faculty seminars; corporate training in Emotional Intelligence and The Art of Delegation.
- **Certifications:** Building Scalable Java Microservices with Spring Boot and Spring Cloud (Google Cloud); AWS Training (Sherdil IT Academy); Scrum Foundation Professional Certificate (SFPC).
- **Community:** volunteer mentor for youth at Al-Khidmat Aghosh Home. **Languages:** English (professional working proficiency), Urdu (native).